Yexiang Cheng

AI Researcher — AI for Scientific Discovery — Foundation Models — LLM Agents

School of Computer and Communication Sciences, EPFL

yexiang.cheng@epfl.ch — LinkedIn — GitHub — Website

Research Summary

AI researcher interested in designing next-generation AI systems for scientific discovery. My work has continuously evolved from explainable deep learning and generative models to foundation models, vision-language models, and LLM-based agent systems, driven by a passion for mastering emerging AI paradigms and translating them into complex, data-driven scientific applications.

Education

EPFL, Master in Computer Science Sep 2024 – Present
GPA: 5.51 / 6.00

Peking University, Bachelor in Bioinformatics Sep 2020 – Jun 2024
GPA: 3.69 / 4.00

Peking University, Bachelor in Computer Science Sep 2020 – Jun 2024
GPA: 3.73 / 4.00

Research Experience

Research Intern, Roche Pharma Research & Early Development May 2026 – Present

- **LLM Agents for Drug Safety Assessment.** Designed an evidence-grounding metric (Evidence Support Rate) for evaluating factual support of agent-generated reports and improved robustness through workflow refinement.
- **Privacy-preserving Long-context QA.** Leading the development of a local agentic framework for confidential clinical reports exceeding 1,000 pages, enabling long-context question answering without commercial LLM APIs.

Research Assistant, AIMM Lab, EPFL Dec 2024 – Apr 2026
Advisor: Prof. Charlotte Bunne

- **Cell-level Representation Learning.** Introduced cell-level representations for spatial proteomics foundation model, preserving cellular, spatial, and marker-interaction information to enable robust downstream analysis beyond handcrafted biological features. Demonstrated the representation on cell phenotyping, cross-panel transfer learning, and biologically meaningful clustering for hypothesis generation and discovery.
- **Foundation Model Benchmarking.** Systematically evaluated the foundation model across diverse downstream tasks spanning cell-, niche-, and tissue-level prediction.
- **Vision-Language Models for Spatial Proteomics.** Investigated spatial reasoning limitations of vision-language models for spatial proteomics and explored lightweight adaptation strategies for improving spatial understanding.

Undergraduate Thesis, School of Life Sciences, Peking University Sep 2023 – May 2024
Advisor: Lect. Fenglin Liu

- **Interpretable Machine Learning for Immunotherapy Response Prediction.** Designed interpretable machine learning models for pan-cancer immunotherapy response prediction using gene expression profiles, leveraging SHAP and LIME to identify predictive biomarkers and improve model interpretability.

Research Assistant, Wangxuan Institute of Computer Technology, Peking University Nov 2022 – May 2023
Advisor: Prof. Jiaying Liu

- **StyleGAN-based Icon Generation.** Designed a dual-head StyleGAN architecture that decomposes edge and color generation for icon synthesis, and designed a low-level discriminator to improve local texture consistency and boundary smoothness.

Research Intern, Helixon Jan 2022 – Jun 2022
Advisor: Prof. Jianzhu Ma

- **Explainable Deep Learning for Retinal Disease Classification** Developed a variational-dropout-based training framework that guides CNNs to focus on expert-annotated retinal lesions, improving model interpretability for macular degeneration classification.

Selected Publications

AI-powered Virtual Tissues from Spatial Proteomics for Clinical Diagnostics and Biomedical Discovery
J. Wenckstern*, E. Jain*, B. von Querfurth*, **Y. Cheng***, K. Vasilev, M. Pariset, P. F. Cheng, P. Liakopoulos, O. Michielin, A. Wicki, G. Gut, C. Bunne.
Nature, accepted, in press.

TSAssistant: A Human-in-the-Loop Agentic Framework for Automated Target Safety Assessment
X. Zheng*, Z. Jiang*, D. Tokar†, **Y. Cheng**†, A. Serra, M. Guerard, K. Hatje, T. Doktorova.
arXiv preprint, 2026.

* Equal contribution † Equal second contribution

Teaching & Service

Teaching Assistant, Introduction to Computation, Peking University Sep 2023 – Jan 2024
Q&A support, assignment grading, and final exam design for C programming.

Teaching Assistant, Bioinformatics Lab, Peking University Feb 2023 – Jun 2023
Assisted students with R, Linux, probability, statistics, and machine learning.

Selected Honors

Excellent Graduate of Beijing, awarded to top 5% students	2024
Excellent Graduate of Peking University, awarded to top 10% students	2024
Qin Wanshun–Jin Yunhui Scholarship, School of Life Sciences, PKU	2020, 2023
Merit Student, Peking University	2023
UBIQUANT Scholarship	2022
The Third Prize of Peking University Scholarship	2021
Gold Medal, China National Biology Olympiad (CNBO)	2019

Technical Skills

AI architecture	Foundation Model, Vision-Language Model, Representation Learning, Benchmarking, Generative Models
LLM Agents & Systems	Multi-Agent Systems, RAG & Long-Context QA, Agent Evaluation Metrics
Scientific AI & Explainability	Spatial Proteomics, Gene Expression Analysis, Interpretable ML
Languages & Tools	Python, C/C++, R, Linux, Git